

VIEREMA KAARAKKALA, Finland

WMO: 028340

Lat: 63.841N

Lon: 27.221E

Elev: 681

StdP: 14.34

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-5.9	-14.6	2.6	-10.4	-9.5	3.4	-5.6	18.6	18.7	16.6	21.0	5.4	100	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	78.7	64.0	75.0	62.3	71.6	60.2	66.7	75.0	64.5	71.9	62.4	68.9	7.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	90.5	69.6	61.6	84.1	67.8	59.4	77.7	65.5	31.7	75.0	29.9	72.2	28.4	68.8	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	14.4	13.2	-12.9	82.1	4.8	5.6	-16.3	86.2	-19.1	89.5	-21.8	92.7	-25.2	96.7		
			-12.8	69.2	6.4	3.1	-17.4	71.4	-21.1	73.2	-24.7	75.0	-29.4	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	13.7	18.0	24.9	35.7	48.3	55.1	61.3	57.5	49.2	36.4	28.9	22.0
	DBStd	17.79	12.09	11.00	7.90	6.63	9.07	7.58	6.86	6.12	6.11	6.50	7.52	10.38
	HDD50	5427	1126	896	777	431	140	33	4	10	85	424	632	869
	HDD65	10036	1591	1316	1242	878	523	309	157	243	473	888	1082	1334
	CDD50	936	0	0	0	3	87	186	354	243	62	1	0	0
	CDD65	69	0	0	0	0	4	12	41	12	0	0	0	0
	CDH74	405	0	0	0	0	32	73	243	57	0	0	0	0
	CDH80	49	0	0	0	0	1	5	38	5	0	0	0	0
	WSAvg	7.0	7.2	7.7	7.8	7.1	7.2	6.6	5.7	6.2	6.3	6.9	7.5	8.0
	PrecAvg	27.6	1.9	1.6	1.4	1.4	2.2	2.9	3.5	3.0	2.5	2.5	2.4	2.3
Precipitation	PrecMax	37.4	3.7	3.1	2.7	2.4	4.4	6.0	6.2	6.1	4.2	4.3	4.6	5.4
	PrecMin	20.4	0.5	0.2	0.1	0.5	0.6	1.1	0.8	1.0	0.8	0.5	0.4	0.7
	PrecStd	4.3	0.8	0.8	0.6	0.5	1.0	1.3	1.4	1.1	1.0	1.1	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	39.3	44.9	62.5	78.0	80.2	83.5	79.7	67.8	53.1	47.2	39.1
		MCWB	35.5	36.5	37.8	48.9	57.7	63.5	66.5	66.9	59.9	49.9	46.0	38.9
	2%	DB	32.8	35.9	40.9	54.1	72.9	75.9	80.1	74.7	63.1	50.0	42.3	36.6
		MCWB	32.6	34.4	35.5	44.1	56.2	61.8	65.8	63.2	56.6	48.1	41.7	36.0
	5%	DB	31.8	33.2	37.3	50.1	68.4	71.4	76.2	70.9	60.0	47.5	39.7	34.4
		MCWB	31.7	32.5	33.7	42.1	54.8	59.0	64.3	61.0	55.0	46.0	39.3	34.0
	10%	DB	30.2	31.8	34.8	46.3	63.0	67.7	72.6	67.3	57.6	45.1	37.7	32.7
		MCWB	30.1	31.5	32.5	39.6	51.2	57.1	62.1	59.6	53.8	43.9	37.4	32.5
	0.4%	WB	35.4	37.2	38.9	50.0	60.7	66.7	70.1	68.7	60.5	51.7	46.0	39.0
		MCDB	35.8	39.0	43.3	58.1	72.6	74.5	78.1	76.9	66.5	52.9	46.7	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.7	34.6	36.5	46.1	58.1	63.8	67.6	64.6	58.3	48.7	41.9	36.2
		MCDB	32.8	35.7	39.3	52.2	69.2	71.7	76.2	71.7	61.7	49.5	42.4	36.4
	5%	WB	31.7	32.7	34.6	42.5	55.4	60.7	65.9	62.5	56.2	46.3	39.4	34.0
		MCDB	31.8	33.3	36.7	48.6	67.3	69.8	73.9	68.3	58.9	47.3	39.7	34.3
	10%	WB	30.1	31.5	32.9	40.0	52.3	58.3	63.9	60.7	54.6	44.0	37.5	32.5
		MCDB	30.2	31.8	34.3	45.0	62.3	66.1	71.0	66.0	57.1	45.1	37.6	32.6
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	11.0	12.4	15.2	14.5	14.0	12.9	10.7	7.8	6.3	7.0
		MCDBR	6.0	8.2	12.3	18.8	20.6	18.5	17.2	16.6	14.0	10.2	7.2	6.6
		MCWBR	6.1	7.6	8.8	11.1	10.2	8.8	7.8	7.8	8.4	8.3	7.3	6.4
	5% WB	MCDBR	5.9	7.9	10.1	17.0	19.1	16.4	15.9	14.4	12.3	9.6	7.0	6.6
		MCWBR	6.0	7.4	8.0	10.7	10.6	8.5	8.3	8.0	8.1	8.7	7.1	6.5
Clear-Sky Solar Irradiance	taub		0.213	0.276	0.303	0.341	0.340	0.340	0.363	0.345	0.323	0.302	0.367	N/A
	taud		1.976	2.124	2.184	2.303	2.463	2.481	2.432	2.471	2.525	2.396	2.697	N/A
	Ebn at Noon		117	201	247	265	278	279	268	263	247	197	109	N/A
	Edh at Noon		9	20	29	32	30	30	31	27	21	16	8	N/A
All-Sky Solar Radiation	RadAvg		50	219	680	1136	1504	1665	1580	1191	661	248	60	19
	RadStd		6	47	73	135	164	120	164	121	71	42	12	2

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page